VACUUM PUMP [SKYACTIV-D 1.5]

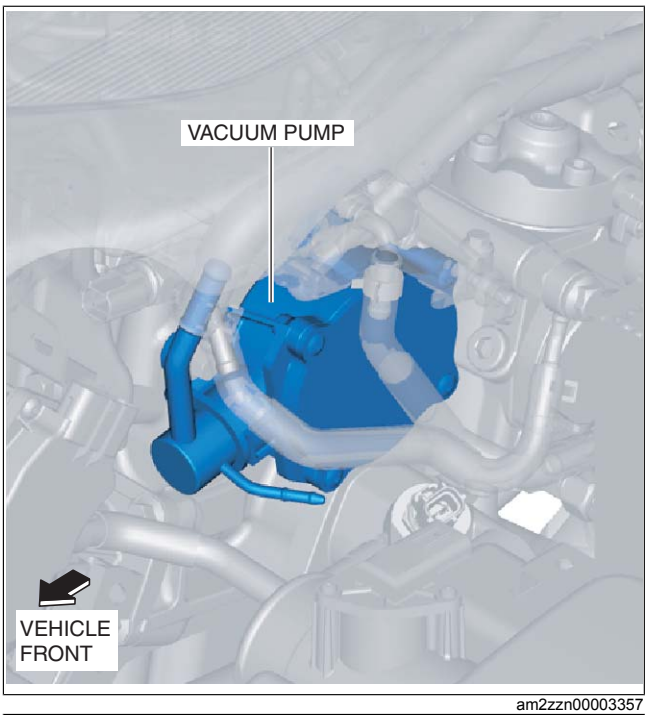
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Purpose, Function

- Ensures vacuum required for braking force to compensate for the decrease in intake manifold vacuum applied to the power brake unit caused by the retard intake valve closing. In addition, it secure vacuum pressure necessary to operate the turbocharger actuator of the turbocharger with variable turbine geometry.

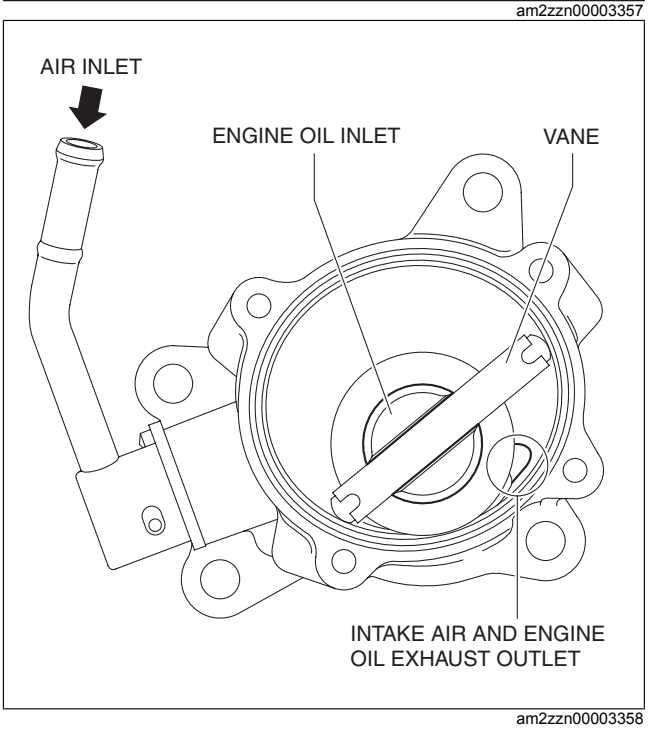
Construction

- The vacuum pump is located on the engine (rear).



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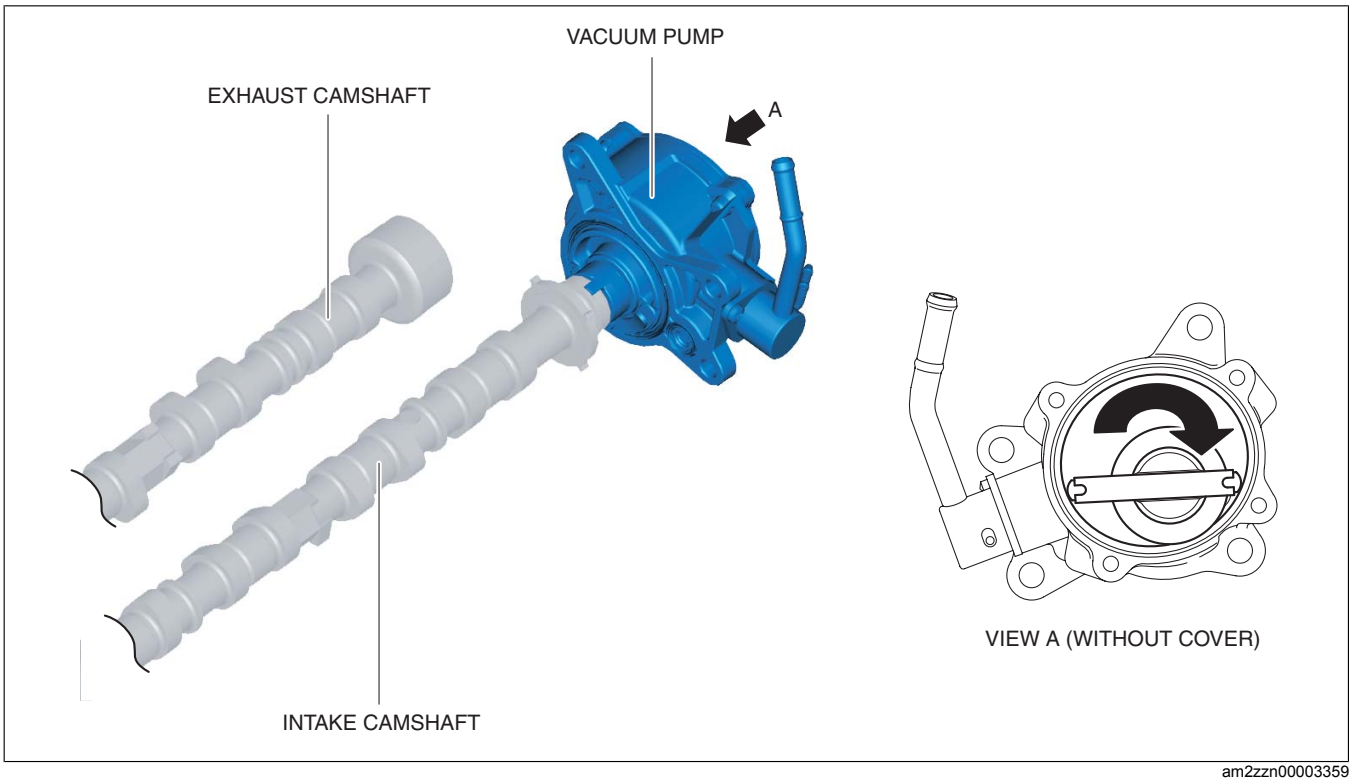
- For smooth vane rotation, a single-vane type vacuum pump equipped with a lubrication passage has been adopted for circulating a part of the engine oil.



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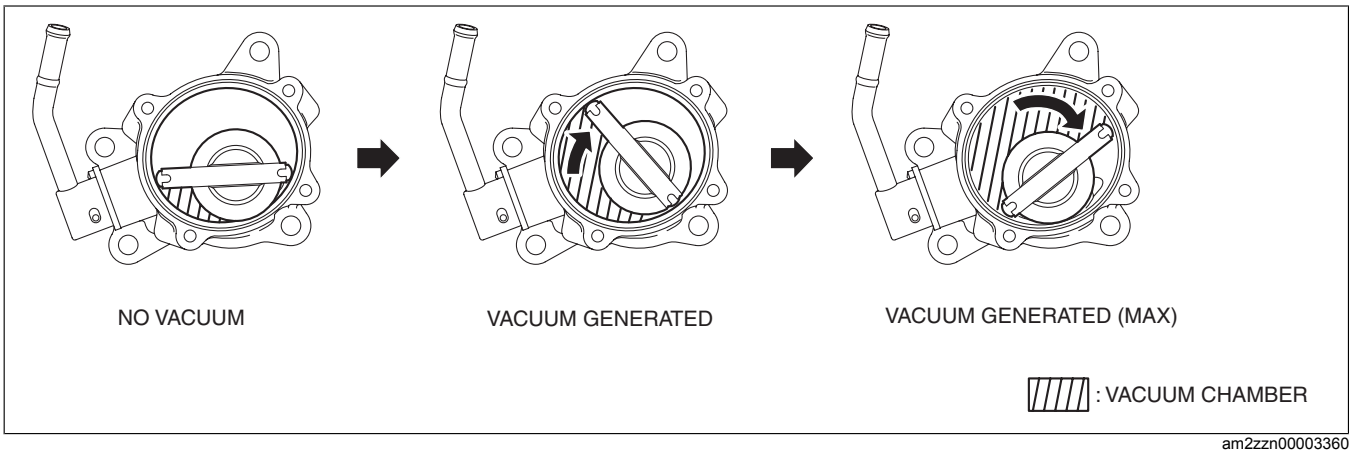
Operation

- Because the vacuum pump is directly engaged with the intake camshaft, the vane in the vacuum pump rotates together with the intake camshaft, and the vacuum chamber is expanded and vacuum is generated. Intake air and engine oil are exhausted from the exhaust outlet.



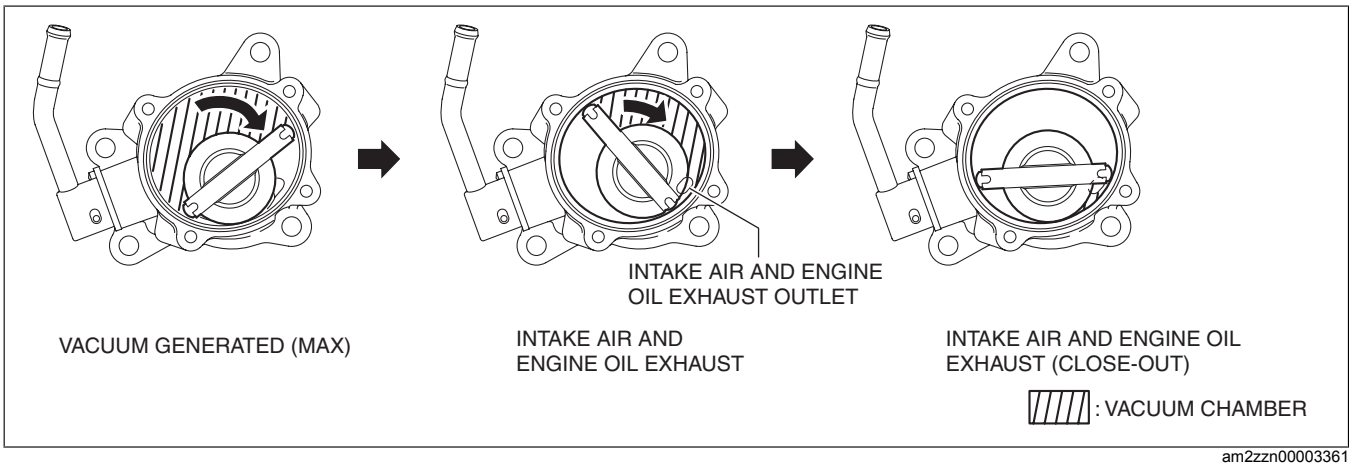
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Vacuum generation mechanism



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Intake air and engine oil exhaust mechanism



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